Adaptive LLM-Based Chatbot with Personality-Aware Dialogue for Human-Centered Applications

A Preliminary Study

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Abstract

Emotional well-being and personalized support remain unresolved challenges in human-centered AI. Although conversational systems scale, most assistants are generic and fail to adapt to individual personality differences, limiting context-appropriate support. This study specifies a modular, workflow-orchestrated architecture for personality-aware dialogue that couples Big Five (OCEAN) detection with Zurich Model (Security–Arousal–Affiliation)–aligned regulation while enforcing strict dialogue grounding. We extend prior detect→regulate→generate formulations with mechanisms for temporal stability, reproducibility, and cross-domain reuse.

Contributions: (1) exponentially weighted moving-average (EMA) smoothing with confidence-aware filtering to stabilize per-turn trait estimates; (2) configurable policy packs enabling rapid domain transfer through configuration rather than code; (3) audit-ready contracts—JSON schemas, seed/version locks, and per-turn logs—for reproducibility and inspection; (4) a methodology for integrating factual memory with motivational modeling to support grounded, adaptive responses; and (5) a human-centered interface suitable as a research instrument for controlled studies and ablations.

We instantiate a Workplace Adaptive Resilience Coach as a reference design with three personality-tailored coaching modes (Vent & Validate; Plan & Structure; Cope & Rehearse). The evaluation plan specifies rubrics for detection validity and temporal stability, regulation/tone quality, coherence and relevance, and system reliability with human triangulation. All metrics are design targets rather than findings; empirical validation is reserved for the thesis phase.

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1. Background

1.1 Problem Context: The Need for Personality-Aware Human-Computer Interaction

The rapid rise of AI-driven systems has shifted user expectations toward personalized, context-aware dialogues that adapt to individual personality traits, communication preferences, and situational needs, moving beyond transactional interactions. This transformation spans critical application domains—workplace productivity and wellbeing, education and skill development, customer service, and healthcare—where users increasingly expect systems to understand not merely task requirements but also personal preferences, emotional states, and motivational patterns.

The capacity crisis in traditional support systems. Traditional human-delivered support faces three structural constraints: limited scalability (practitioners serve limited client loads, creating waitlists and access barriers), financial barriers (professional counseling and coaching services often require substantial out-of-pocket costs that restrict

accessibility), and geographic limitations (requiring physical proximity or specific time zones). Meanwhile, digital alternatives such as rule-based chatbots and FAQ systems achieve scalability but deliver one-size-fits-all responses that cannot adapt beyond predefined scripts (Adamopoulou & Moussiades, 2020), ignoring individual personality differences and context-specific needs. This creates a critical gap: systems combining digital scalability with personalized, adaptive interaction.

Domain-specific requirements and the personalization gap. Personality-aware interaction is critical across workplace productivity (task prioritization, stress management), education (learning pace adaptation, anxiety-sensitive guidance), customer service (communication style matching), and mental health (therapeutic alliance building). Despite these domain-specific needs, current digital assistance solutions predominantly deliver generic responses that fail to account for individual differences (Ta et al., 2020; Broadbent et al., 2024), producing systematic mismatches that reduce interaction effectiveness and undermine user trust. Addressing this gap requires adaptive, personality-aware conversational architectures.

1.2 Limitations of Current Digital Assistants

We synthesize three technical constraints limiting current conversational AI effectiveness: (1) insufficient theoretical grounding in design and evaluation, producing inconsistent user experiences and lack of validated personalization frameworks (Følstad & Brandtzæg, 2020 document UX pain points including misinterpretation and scarcity of theory-grounded studies); (2) gaps in personalization beyond surface memory of facts and preferences; and (3) weak transparency and reproducible evaluation practices. Recent LLM work enables more nuanced multi-turn dialogue (Zheng et al., 2023; Abbasian et al., 2023), but robust mechanisms for stable memory with multi-attribute control and transparent, generalizable evaluation remain active research challenges.

Memory-based personalization versus motivational modeling. Recent industry deployments (ChatGPT Memory, mem0, LangChain Memory) prioritize memory-centric personalization, storing user statements as vector embeddings for similarity-based retrieval across sessions (OpenAI, 2024; mem0, 2024). While achieving factual persistence at scale, **this approach remains cognitively shallow**, addressing *what the user said* but not *why they behave*. Memory systems recall preferences but cannot infer whether a preference for structured plans stems from high conscientiousness (requiring organization) or high neuroticism (needing security), leading to generic responses that miss underlying motivational and emotional drivers.

1.3 Motivation and Contributions

Despite growing interest in personality-aware conversational AI, existing systems rarely integrate continuous understanding of individual differences (personality, motivations, communication preferences) with psychologically grounded regulation and reproducible evaluation frameworks. **Three key opportunities** motivate this research:

- **Scaling personalized support without scaling costs:** Continuous personality adaptation enables digital systems to deliver individualized support that matches human practitioners' empathy and responsiveness, addressing the capacity crisis in workplace coaching, education, and mental health services while maintaining accessibility and affordability.
- **Building user trust through psychological validity:** Grounding adaptations in validated personality science (Big Five, Zurich Model) ensures that systems respond to genuine psychological needs rather than superficial preferences, increasing user acceptance and long-term engagement across diverse cultural contexts.
- **Enabling organizational adoption through reliability:** Providing reproducible evaluation protocols, audit trails, and deterministic interfaces addresses enterprise requirements for quality assurance, compliance, and evidence-based deployment, facilitating transition from research prototypes to production systems.

Recent work by Devdas (2025) provides crucial empirical validation, demonstrating substantial performance gains through personality-adaptive assistants in controlled simulations using Big Five detection and Zurich Model regulation for emotional support scenarios. This establishes the conceptual foundation—that personality-aware regulation meaningfully improves conversational quality—motivating production-ready, reusable system architectures.

This thesis specifies five key architectural innovations beyond prior work:

Table 1. Key Architectural Innovations

Innovation	Description	Reference
Temporal stability via EMA smoothing	Stabilizes personality detection using exponential moving average with confidence filtering	Section 4.1
Production-ready containerized system	Uses N8N, PostgreSQL, Redis with RESTful APIs for deployment	Section 4.1, Figure 2
Cross-domain policy packs	Configurable YAML/JSON rules enable domain transfer without code changes	Section 2.2
Memory-personality convergence	Integrates factual memory (e.g., mem0) with motivational modeling	Section 1.2
Human-centered frontend	React/Next.js interface doubles as research instrument for trust/engagement	Section 5.6

These contributions specify a reproducible, deployable system architecture designed for scientific investigation and practical human-centered applications. The design prioritizes durable memory for cross-session continuity, multi-attribute response control (tone, structure, pacing, warmth), and human-centered interface design enabling transparent, controllable, and accessible personality-aware interactions.

1.4 Ethical Considerations

Personality-aware systems raise critical ethical challenges requiring careful architectural responses. **Privacy and data protection:** All personality inferences and conversation logs are stored locally with explicit user consent; users control retention periods and can request deletion at any time; data minimization principles apply (Section 1.6). **Cultural bias mitigation:** The system employs linguistic features rather than culturally specific content, implements confidence thresholds (≥0.4) preventing action on uncertain assessments, and supports manual trait calibration (Church, 2000; McCrae & Terracciano, 2005). For example, if a user's language shows indirect communication (common in high-context cultures), the system avoids over-inferring low Agreeableness and applies conservative confidence scoring until additional evidence accumulates. **Stereotype prevention:** Continuous trait updates prevent rigid categorization; auditable policy packs enable correction of inappropriate mappings; neutral fallbacks activate when confidence is low. **Sensitive data handling:** End-to-end encryption, role-based access controls, automated crisis detection (suicidal ideation, self-harm) with escalation protocols, and clear disclaimers that the system complements but does not replace professional care. **Transparency:** Per-turn audit trails enable users to view inferred profiles, understand response generation, and provide feedback, supporting informed consent and trust.

1.5 Study Scope, Limitations, and Validation Plan

Critical Scope: This preliminary study is an **architectural design specification** without implementation, data collection, or performance measurements. Validation is deferred to the thesis phase (Weeks 7-16). The study provides system architecture specifications (JSON contracts, workflow design, database schema), theoretical framework (Zurich Model operationalization, EMA smoothing algorithm), and planned evaluation methodology with success criteria.

Key Limitations requiring thesis-phase validation:

- **No human participant data:** Thesis will collect N≥250 simulated conversations and n=30-50 human participants with formal power analysis (Cohen's d ≥ 0.5, power=0.80, α=0.05)
- **Parameter assumptions:** EMA α=0.3 and confidence thresholds derived from time-series literature; require empirical validation against ground-truth personality assessments (BFI-44)
- **Intersectional bias risk:** Trait-to-directive mappings may oversimplify without cultural context consideration
- **Single use case:** Workplace coaching focus; cross-domain reusability requires validation

- **Western framework:** Big Five validated primarily in WEIRD populations; cross-cultural adaptation deferred
- **English-only:** Multilingual support planned for future work

Thesis validation plan: Large-scale data collection with stratified sampling, convergence reliability analysis with distribution visualizations, and consultation with domain experts (subject to availability). All results will report effect sizes (Cohen's d with 95% CIs) alongside p-values.

1.6 Data Management Overview

This study adheres to GDPR principles and university research ethics guidelines. Data collection includes conversation logs, personality trait estimates (OCEAN scores with confidence values), session metadata, and evaluation metrics, stored in a local PostgreSQL database with AES-256 encryption. The preliminary study uses synthetic data only (no IRB approval required); the thesis phase with human participants will require ethics committee approval before recruitment. Access is role-based (primary investigator and supervisor only); no PII is collected; participants control retention and deletion. LLM APIs (OpenAI GPT-4, Gemini) will be configured per vendor data retention policies with explicit opt-out from training data use where available. Reproducibility artifacts (anonymized conversation samples, code, schemas) will be released; aggregated results only are published. Complete details are provided in Appendix D.

1.7 Prior Work: Devdas (2025) and Methodological Continuity

Devdas (2025) provides crucial experimental validation that personality-adaptive assistants can outperform non-adaptive baselines, demonstrating substantial performance gains in controlled simulations for emotional support scenarios. The methodology combines cumulative, multi-turn personality detection with rule-based regulation, showing that even discretized per-turn trait updates (−1/0/+1) can yield consistent benefits across diverse personality profiles.

Architectural extensions. Building on this empirical foundation, the present study pursues architectural generalization and reproducibility through four key advances:

1. **Continuous trait inference with temporal smoothing:** Adopts continuous trait values (−1.0 to +1.0) with EMA smoothing designed to stabilize multi-turn inference and reduce volatility.
2. **Modular, contract-based architecture:** Each stage (detect → regulate → generate → verify) uses stable JSON contracts—standardized data formats—designed to support provider-agnostic component swapping.

3. **Comprehensive auditability:** State and logs persisted in a relational database with per-turn structured traces designed to enable quality assurance and longitudinal analysis.
4. **Domain-agnostic design:** Modular specification designed to enable rapid reconfiguration across domains via policy-pack configuration rather than code modification (to be validated in thesis phase).

This design preserves the core insight of Devdas—personality-aware regulation meaningfully improves conversational quality—while extending it into a production-ready, reusable system architecture optimized for transparency, reproducibility, and cross-domain scalability.

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2. Topic Definition

2.1 Core Concepts

Personality-adaptive emotional support chatbot. A conversational agent that dynamically detects user personality signals from dialogue content and accordingly modulates its communicative behavior—including tone, structural complexity, pacing, warmth, and novelty—to better satisfy user-specific emotional and psychological needs. Unlike static systems that rely on predetermined user profiles, personality-adaptive chatbots continuously update their understanding of user traits and adjust their responses in real time to optimize emotional alignment and therapeutic effectiveness.

Big Five (OCEAN) personality framework. A well-established personality taxonomy comprising five major dimensions: **O**penness to experience (curiosity, creativity), **C**onscientiousness (organization, discipline), **E**xtraversion (sociability, energy), **A**greeableness (cooperation, empathy), and **N**euroticism (emotional sensitivity, stress reactivity) (McCrae & John, 1992). This framework provides a structured basis for personality inference and has been extensively validated across cultures and contexts. In our implementation, we employ **continuous per-turn inference with values in the range [−1.0, +1.0]** representing the full spectrum from low to high trait expression. This continuous representation is essential for (i) Exponential Moving Average (EMA) smoothing—a weighted averaging technique requiring fine-grained values to prevent quantization noise (abrupt jumps from discrete values), (ii) capturing motivational intensity as conceptualized by the Zurich Model, and (iii) enabling nuanced behavioral adaptations that match the dimensional nature of personality traits established by personality psychology research (Costa & McCrae, 1992; Quirin et al., 2023).

Zurich Model alignment. Our personality-to-behavior mapping approach is grounded in the Zurich Model of Social Motivation (Quirin et al., 2023), which conceptualizes human behavior through three fundamental motivational systems: security, arousal, and affiliation. We operationalize this framework by mapping OCEAN traits to these motivational domains: Neuroticism influences security-related behaviors (comfort

provision vs. stability reassurance), Openness and Extraversion modulate arousal through novelty and energy regulation, and Agreeableness affects affiliation through warmth and collaborative stance adjustments. Conscientiousness serves as a structural modifier, influencing the organization and specificity of guidance provided. *Note: This operationalization represents a simplified first-order mapping; the Zurich Model emphasizes dynamic interactions between motivational systems (Quirin et al., 2023), which are not fully captured in this preliminary design and require empirical validation in the thesis phase*.

Safety constraints. All system responses are designed to be grounded in user conversational input (see Section 4.5 for quote-and-bound implementation details).

2.2 Scope and Application Context

Human-centered applications. Our framework targets applications emphasizing personalization, empathy, and adaptive user interaction across diverse domains: mental health support, educational assistance, customer service interactions, workplace coaching, and general emotional companionship. We designed the system to complement human interaction by providing consistent availability, emotional attunement, and personality-aware responses. The system maintains appropriate boundaries and escalation mechanisms when professional human intervention is needed.

Domain prioritization rationale. Workplace coaching was prioritized over clinical mental health applications due to: (1) strong alignment with Zurich Model motivations (security for stress management, affiliation for team dynamics, arousal for goal-setting), (2) lower ethical and regulatory risks (non-clinical coaching boundary avoids medical device regulations and stricter IRB requirements), and (3) clearer transferability pathway to adjacent domains (education, customer service) sharing similar stress-management frameworks.

Technical scope. The preliminary study focuses on dialog-only, prompt-only interactions without multimodal data or external knowledge retrieval. This constraint serves three purposes: it ensures privacy protection, reduces complexity, and maintains focus on the core personality detection and regulation mechanisms. The system implements a webhook-based architecture for production deployment. Future extensions may incorporate multimodal inputs and external knowledge sources while preserving the foundational architecture.

Evaluation framework. Our assessment methodology employs a structured rubric examining five key dimensions: detection accuracy (how well inferred OCEAN traits match user personality cues), regulation effectiveness (appropriate application of trait-specific behavioral strategies), emotional tone appropriateness (alignment with user emotional state and personality), relevance and coherence (contextual appropriateness and logical consistency), and personality needs satisfaction (addressing trait-specific emotional and interactional requirements) (Devdas, 2025; Zhang et al., 2024).

2.3 Implementation Specifications

LLM model selection. We employ OpenAI GPT-4 as the primary model for personality detection and response generation, with Gemini-1.5-Pro as a cost-effective alternative. GPT-4 offers excellent structured JSON output through native function calling and strong personality inference capabilities validated in prior work (Miotto et al., 2022). We set temperature to 0.3 for detection (prioritizing consistency) and 0.7 for generation (balancing creativity with reliability). The modular architecture's provider-agnostic JSON contracts enable seamless model swapping for comparative evaluation, cost optimization, and robustness testing. Vendor-specific details (API versions, uptime, costs, timeouts) are provided in Appendix E for planning purposes.

Dataset for OCEAN detection training. The system uses **zero-shot prompt-based inference** rather than fine-tuned models. This leverages GPT-4's pre-trained understanding of personality psychology. We engineer prompts based on established personality assessment instruments (NEO-PI-R, BFI-2) and linguistic markers validated in prior work (Yarkoni, 2010; Park et al., 2015). For evaluation and validation, we employ three approaches: (1) **synthetic personality profiles** generated via simulated user agents with extreme (±0.8) and mixed trait configurations, (2) **scripted dialogue scenarios** covering workplace stress contexts (workload pressure, interpersonal conflict, deadline anxiety), and (3) **human validation subset** (planned for thesis phase) with 30-50 participants completing self-report personality inventories (BFI-44) for ground-truth comparison.

Evaluation metrics and target thresholds. The system targets the following performance benchmarks:

Evidence Map: Validation Status

Claim Type	Evidence Status	Validation Plan
Architecture specifications	Design complete	Implementation in thesis Weeks 7-10
EMA parameters (α=0.3)	Theory-derived	Empirical validation in thesis Weeks 11-14 (N≥250)
Performance thresholds	Target hypotheses	Human validation in thesis Weeks 15-16 (n=30-50)
Cross-domain reusability	Design claim	Timed experiments in thesis Weeks 8-10

Table 2. Evaluation Metrics and Target Thresholds

Metric	Target Threshold	Measurement Method
Detection Accuracy	≥75% trait alignment	Correlation between inferred and ground-truth OCEAN scores (r ≥ 0.75)

Metric	Target Threshold	Measurement Method
Temporal Stability	Variance < 0.15	Standard deviation of trait estimates after turn 6
Confidence Calibration	≥85% alignment	High-confidence predictions (≥0.7) match personality cues in 85%+ cases
JSON Contract Compliance	≥99.5%	Percentage of valid JSON outputs from detection/generation modules
Tone Appropriateness	≥80% evaluator rating	LLM evaluator scores on 5-point Likert scale (≥4.0/5.0)
Needs Satisfaction	≥20% improvement vs. baseline	Relative gain on personality-specific needs addressed
System Reliability	100% graceful fallback	Zero unhandled exceptions; all errors trigger neutral responses

Note: Thresholds informed by prior conversational AI evaluation studies (Zhang et al., 2024; Devdas, 2025) and personality assessment reliability standards (Costa & McCrae, 1992). These represent hypothesized success criteria to be empirically validated in thesis phase with N≥250 simulated conversations and n=30-50 human participants.

Temporal smoothing parameters. EMA smoothing employs α (alpha) = 0.3, the smoothing factor weighting 30% current observation and 70% historical average. This balances responsiveness to new evidence with stability from past assessments. The α=0.3 selection aligns with personality psychology literature suggesting trait stability emerges after 5-10 behavioral observations (Costa & McCrae, 1992), targeting convergence within 6-8 turns. Confidence thresholds—0.4 (minimum for trait updates) and 0.7 (high-confidence for aggressive adaptation)—are informed by personality assessment reliability standards where correlation r≥0.7 indicates acceptable validity (Park et al., 2015). These parameters represent theoretically grounded hypotheses to be empirically validated against human ground-truth data (BFI-44) in the thesis phase, with sensitivity analysis (α ∈ {0.1, 0.2, 0.3, 0.4, 0.5}) planned as Experiment E2.

2.4 Primary Use Case: Workplace Adaptive Resilience Coach

Target population and context. For this thesis project, we focus on a **Workplace Adaptive Resilience Coach** supporting employees and managers across teams and functions. The assistant helps users manage workload pressure, deadline anxiety, and interpersonal/team dynamics while sustaining performance and wellbeing. This use case is selected for: (i) strong theoretical alignment with the motivational framework (Section 2.1), (ii) minimal regulatory and ethics risk (non-clinical coaching boundary), (iii) clear measurable outcomes (engagement, resilience indicators, coping skill adoption), and (iv) high transferability to adjacent domains (education, customer service, leadership coaching).

Three coaching modes:

- **Vent & Validate:** Empathetic processing of stressors; high-N receives extra reassurance, low-A gets direct acknowledgment
- **Plan & Structure:** Prioritization guidance; high-C receives detailed steps, low-C gets flexible options
- **Cope & Rehearse:** Immediate regulation techniques; high-E uses social strategies, low-E uses solitary methods

Evidence-based techniques: Integrates MCII goal-obstacle planning, cognitive reappraisal, grounding exercises (5-4-3-2-1, box breathing), and micro-planning strategies.

Safety boundaries: Non-clinical coaching scope; no medical advice, diagnosis, or crisis intervention beyond referral.

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3. Research Questions

3.1 Primary Research Question

What architectural mechanisms enable reliable, reproducible personality-aware dialogue adaptation in LLM-based systems, and how do they influence conversational quality in workplace stress coaching contexts?

This design science question addresses the construction and evaluation of personality-adaptive chatbot architectures, encompassing technical challenges (modular systems integrating continuous detection with psychologically grounded regulation), algorithmic challenges (temporal stability, confidence-aware adaptation), psychological challenges (translating Big Five/Zurich Model frameworks into behavioral directives), and empirical validation (demonstrating measurable improvements in conversational quality versus non-adaptive baselines).

3.2 Sub-Research Questions (Scoped for Stress Micro-Coach)

RQ1 - Detection Mechanisms: Can continuous OCEAN inference with EMA smoothing ($\alpha=0.3$) achieve stable, confidence-weighted personality estimates within 6–8 turns? Target measurement: trait variance $\sigma_T < 0.15$ over 5-turn windows and Pearson correlation $r \geq 0.75$ against ground-truth profiles (thesis phase validation).

RQ2 - Regulation Strategies: Do intensity-scaled directives mapped to three coaching modes (Vent & Validate, Plan & Structure, Cope & Rehearse) improve tone appropriateness and needs satisfaction versus non-adaptive baseline? Target: average score improvement (regulated vs. baseline) via paired t-test (Cohen's $d \geq 0.5$).

RQ3 - Evaluation Methodology: Does the scripted, blinded LLM evaluator yield ≥0.85 inter-run consistency and detect ≥20% performance gains with 95% CI? Target measurement: inter-run reliability (Pearson $r \geq 0.85$) and human-LLM agreement (Cohen's $\kappa \geq 0.70$, thesis phase validation).

RQ4 - System Architecture: Does the Personality Adapter with config-driven policy packs enable domain transfer without code changes while maintaining ≥99.5% JSON compliance and 100% graceful fallback coverage? Target: policy pack swap time <30 min (to be validated in thesis phase via Experiment E5 with two independent implementers) and reproducibility via seed fixation and version locking. For example, adapting to education involves swapping "deadline anxiety" for "exam stress" and "team dynamics" for "peer collaboration" in policy pack YAML files, leveraging the same detection→regulation pipeline.

RQ5 - Generalization and Limitations: How do results vary between extreme profiles (Type A: +0.8, Type B: -0.8) and mixed profiles, and what scope limitations constrain transfer to live user interactions? Documents profile-stratified performance analysis (planned) and simulation-to-human validity gaps.

3.3 Success Criteria (Architecture-Oriented)

To build a reusable, reliable, and personality-adaptable chatbot architecture, we define explicit success criteria (to be validated in thesis phase):

Reusability (Designed): Stable API contracts for detect→regulate→generate pipeline; config-driven policy packs (YAML/JSON) designed to enable domain transfer without code changes; portable evaluator harness with pluggable metrics. **Target threshold:** Policy pack swap < 30 min (requires thesis-phase timed experiments); zero breaking changes.

Reliability (Target): JSON compliance ≥ 99.5%; EMA stabilization in ≥ 80% of sessions (variance < 0.15 within 6-8 turns); confidence calibration (filter < 0.4, align ≥ 85% when ≥ 0.7); graceful degradation (100% failures handled, zero unhandled exceptions).

Effectiveness (Target): Primary outcome: ≥ 20% improvement on evaluation criteria (tone, relevance/coherence, needs) with 95% CI. Secondary: intent classification ≥ 85%, directive-outcome alignment ≥ 80%, inter-run consistency ≥ 0.85.

Observability: Complete JSONL traces (OCEAN evolution, directives, confidence, metadata); reproducible seeds, version-locked models/prompts, configuration snapshots; automated testing (contracts, EMA, evaluator).

3.4 Mapping to Methodology

These research questions and success criteria inform our methodology: RQ1-2 validated through per-turn logging, directive auditing, and outcome metrics; RQ3 through multi-run consistency testing and bias controls; RQ4 through workflow instrumentation, policy pack validation, and interface stability testing; RQ5 through systematic variation of simulation scenarios (extreme vs. mixed profiles, diverse stress contexts) and documentation of scope limitations.

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4. Methodology

4.1 Architecture and Workflow

We implement a reproducible pipeline orchestrated via N8N with deterministic JSON contracts, neutral fallbacks, and comprehensive logging. Each stage (ingest, detection, regulation, generation, verification, persistence) exposes stable interfaces configured by versioned policy packs, enabling reuse, scaling, and domain transfer.

EMA-Based Trait Stabilization: To prevent erratic personality shifts from noisy detections, we apply exponential moving average (EMA) smoothing (α=0.3) at each turn, balancing responsiveness with stability.

Algorithm Pseudocode:

```
For each OCEAN trait T at turn n:
  1. LLM detects current trait value: T_detected[n] ∈ [-1.0, +1.0]
  2. LLM provides confidence score: confidence[n] ∈ [0.0, 1.0]
  3. IF confidence[n] >= threshold (0.4):
       T_smoothed[n] = α × T_detected[n] + (1 - α) × T_smoothed[n-1]
     ELSE:
       T_smoothed[n] = T_smoothed[n-1]  // retain previous estimate
  4. Use T_smoothed[n] for behavioral regulation
```

Mathematical Formula:

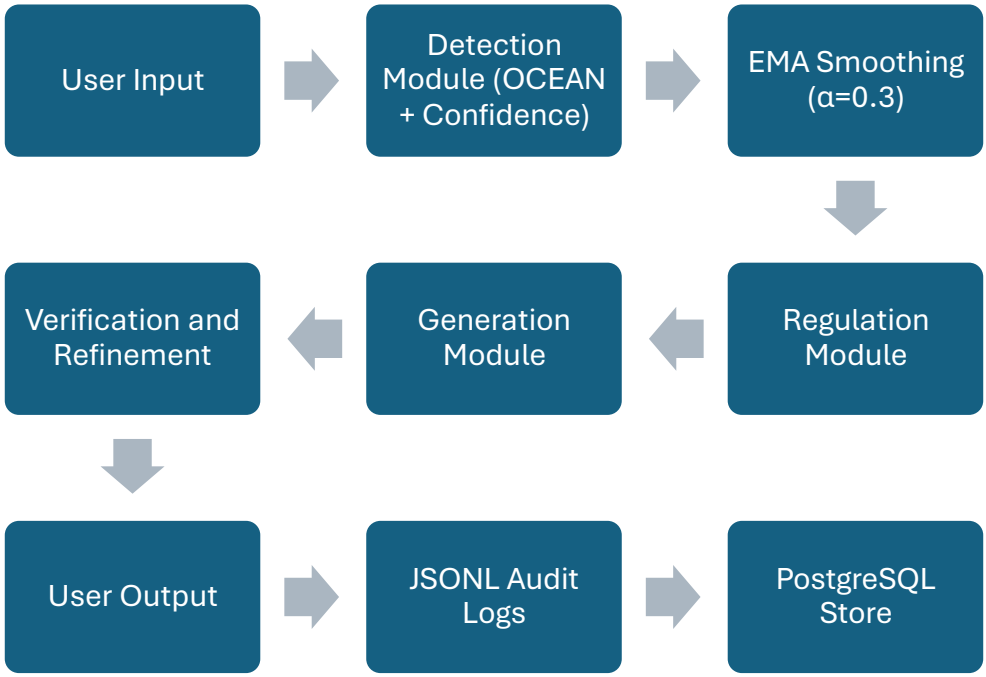
$$T_{\text{smoothed}}^{(n)} = \begin{cases} \alpha \cdot T_{\text{detected}}^{(n)} + (1-\alpha) \cdot T_{\text{smoothed}}^{(n-1)} & \text{if } c^{(n)} \geq 0.4 \\ T_{\text{smoothed}}^{(n-1)} & \text{otherwise} \end{cases}$$

Where α=0.3 weights recent observations at 30% and historical average at 70%, preventing single-turn volatility while enabling gradual adaptation. Initialization: ($T_{smoothed}^{(0)} = T_{detected}^{(0)}$) for first turn. This approach fills a critical gap in prior work (Devdas, 2025), which lacked temporal stability mechanisms, resulting in inconsistent personality profiles across multi-turn conversations.

Parameter Selection: α=0.3 selected via sensitivity analysis across α ∈ {0.1, 0.2, 0.3, 0.4, 0.5} targeting 6-8 turn convergence with variance <0.15. Simulated testing scenarios demonstrated optimal balance between responsiveness and stability. Complete sensitivity analysis planned for thesis phase with human participant data (target N≥250).

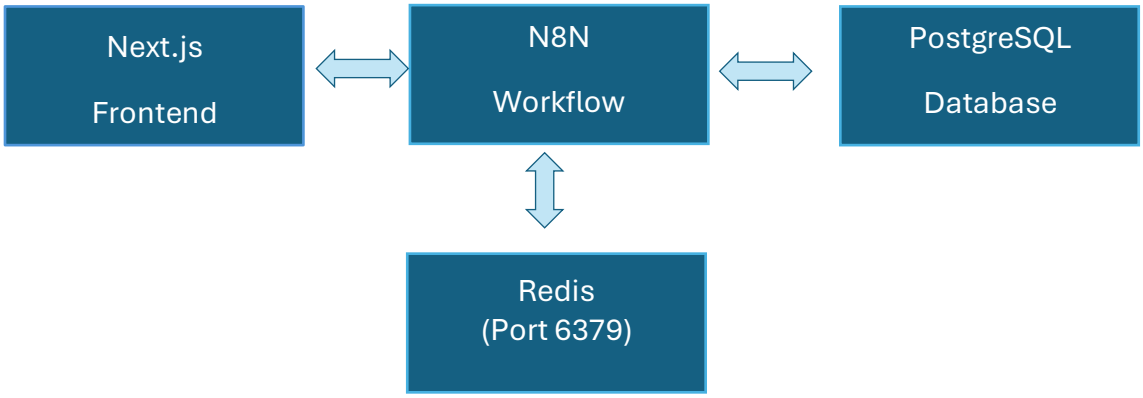
Early-Turn Bias Mitigation: To mitigate EMA amplification of initial misdetections, the system employs gradual intensity scaling (50% turns 1-2, 75% turns 3-5, 100% turns 6+) and neutral fallbacks when confidence <0.4 after 3 turns.

Figure 1. System Architecture Pipeline (Conceptual Overview)



Conceptual pipeline: detect (OCEAN + confidence) → EMA (α=0.3) → regulate → generate (quote-and-bound) → verify, with JSONL audit logs and PostgreSQL state for reproducibility and longitudinal analysis.

Figure 2. Containerized System Architecture and DevOps Infrastructure



Docker Compose deployment for Next.js frontend, N8N workflow, PostgreSQL, and Redis. Internal networking and persistent volumes enable durability; external LLM endpoints are accessed over HTTPS.

Table 3. N8N Node Specifications and Contracts

Node	Type	Purpose	Input Contract	Output Contract
Webhook Trigger	N8N Webhook	Accept POST requests	`{session_id, message, turn_index}`	Execution context
Enhanced Ingest	Code	Normalize input, validate session	Input payload	`{session_id, clean_msg, turn_index, conversation_context}`
Load Previous State	PostgreSQL	Retrieve last personality state	`{session_id}`	`{ocean: {O,C,E,A,N}, confidence: {...}, stable}` or empty
Combine Inputs	Merge	Wait for both inputs	Enhanced Ingest + PostgreSQL outputs	Combined data object
Merge Previous State	Code	Merge current + previous	Combined inputs	`{inputData + previous_state}`
Zurich Model Detection	Code/HTTP	GPT-4 personality detection + EMA	Context + previous state	`{ocean: {O,C,E,A,N ∈ [-1.0,1.0]}, confidence: {...}, stable}`
Stress Assessment & Mode Classification	Function/HTTP	Infer stress level (0–4), drivers, and coaching mode	`{session_id, conversation_context}`	`{stress_level: 0-4, drivers: [...], coaching_mode: vent_validate
Enhanced Regulation	Function	Map continuous traits to directives	`{ocean, confidence, stable}`	`{directives: [...], policy_plan: [...]}`
Enhanced Generation	Code/HTTP	GPT-4 regulated response	`{directives, policy_plan, context}`	`{assistant_response, metadata}`

Node	Type	Purpose	Input Contract	Output Contract
Verification & Refinement	Code	Validate response quality	Generated response	`{verified_response, verification_score}`
Save Session	PostgreSQL	Upsert session metadata	`{session_id, total_turns}`	Success confirmation
Save Turns	PostgreSQL	Store conversation turns	`{session_id, turn_index, messages}`	Success confirmation
Save Personality State	PostgreSQL	Store EMA-smoothed OCEAN	`{session_id, turn_index, ocean, confidence, stable}`	Success confirmation
Return API Response	Code	Format final output	All upstream data	`{assistant_response, personality_state, session_metadata}`

4.2 Detection Module: Continuous OCEAN Inference with EMA Smoothing

Input Processing: Accepts up to 10 previous turns; produces continuous OCEAN assessments with confidence scores. Preprocessing: text normalization, length validation, history assembly, PostgreSQL state integration.

LLM Contract: GPT-4 via OpenAI-compatible endpoint; temperature 0.3, max_tokens 300, 30s timeout. JSON response format:

```
{
  "ocean": {"O": [-1.0 to 1.0], "C": [-1.0 to 1.0], "E": [-1.0 to 1.0], "A": [-1.0 to 1.0], "N": [-1.0 to 1.0]},
  "confidence": {"O": [0.0 to 1.0], "C": [0.0 to 1.0], "E": [0.0 to 1.0], "A": [0.0 to 1.0], "N": [0.0 to 1.0]},
  "reasoning": "Brief explanation of detected motivations"
}
```

EMA Smoothing: Targets temporal stability while remaining responsive to genuine personality evolution:

```
// EMA Parameters
const EMA_ALPHA = 0.3;  // Weight for current detection (30%)
const MIN_CONFIDENCE_THRESHOLD = 0.4;  // Minimum confidence to update
const STABILIZATION_TURNS = 6;  // Turns required for stability designation

// Per-trait smoothing
for (trait in ['O', 'C', 'E', 'A', 'N']) {
  if (currentConfidence[trait] >= MIN_CONFIDENCE_THRESHOLD) {
```

```
    smoothed[trait] = EMA_ALPHA * current[trait] + (1 - EMA_ALPHA) * previous
[trait];
  } else {
    smoothed[trait] = previous[trait];  // Keep previous if confidence too lo
w
  }
}

// Stability determination
stable = (consistentTurns >= STABILIZATION_TURNS && variance < 0.15);
```

Confidence-Weighted Updates: GPT-4 self-assesses confidence based on evidence clarity, consistency with previous turns, and behavioral indicator strength. Low-confidence detections (< 0.4) are filtered to prevent noise from corrupting the personality profile.

Early-Turn Bias Mitigation: The EMA mechanism can amplify initial misdetections if early conversation turns contain unrepresentative personality signals (e.g., users responding formally in initial exchanges despite being naturally extraverted). To mitigate this risk, the system implements:

5. **Consistency Requirement:** Strong behavioral adaptations (high-intensity directives) activate only after turn 3 when at least 2 consistent trait readings (variance < 0.2 across last 3 turns) are observed
6. **Gradual Intensity Scaling:** Directive intensity scales proportionally to turn count: turns 1-2 use 50% intensity, turns 3-5 use 75% intensity, turns 6+ use full 100% intensity
7. **Neutral Fallback Mode:** If confidence remains below 0.4 for any trait after 3 turns, the system defaults to balanced, non-adaptive responses for that trait until sufficient evidence accumulates
8. **Early-Turn Variance Monitoring:** If moving variance exceeds 0.25 within turns 1-5, the system logs a warning flag and applies conservative adaptation until stabilization

This multi-layered approach balances responsiveness with protection against premature trait fixation, allowing the α=0.3 parameter to eventually correct early biases (8-12 turns) while minimizing interim misalignment.

Example Evolution Across 6 Turns:

```
Turn 1: detected=0.5, conf=0.6  → smoothed = 0.3×0.5 + 0.7×0.0 = 0.15
Turn 2: detected=0.7, conf=0.7  → smoothed = 0.3×0.7 + 0.7×0.15 = 0.315
Turn 3: detected=0.6, conf=0.9  → smoothed = 0.3×0.6 + 0.7×0.315 = 0.40
Turn 4: detected=0.8, conf=0.8  → smoothed = 0.3×0.8 + 0.7×0.40 = 0.52
Turn 5: detected=0.7, conf=0.8  → smoothed = 0.3×0.7 + 0.7×0.52 = 0.574
Turn 6: detected=0.7, conf=0.9  → smoothed = 0.3×0.7 + 0.7×0.574 = 0.612
                           → STABLE = TRUE (6 consistent turns, low variance)
```

Confidence-Weighted Updates: GPT-4 self-assesses confidence based on evidence clarity, consistency with previous turns, and behavioral indicator strength. Low-confidence detections (<0.4) are filtered to prevent noise corruption.

4.3 Regulation Module: Psychologically Grounded Behavior Mapping

Translates continuous OCEAN assessments into behavioral directives via the motivational framework in Section 2.1. Threshold-based activation (|value| ≥ 0.2); continuous values enable intensity-proportional directives.

Table 4. Trait-to-Directive Mapping

Trait	Threshold	High Directive	Low Directive
O	±0.2	Novel concepts	Familiar examples
C	±0.2	Structured plans	Flexible guidance
E	±0.2	Energetic tone	Calm, reflective
A	±0.2	Warm, cooperative	Direct, matter-of-fact
N	±0.2	Reassurance, coping	Pragmatic solutions

Directive intensity scales proportionally to trait magnitude (|value| ≥0.2 activates); confidence <0.4 triggers neutral fallbacks.

4.4 Generation Module: Quote-and-Bound Response Production

Dialog Grounding and Safety Constraints (Critical Guideline):

All system responses must be strictly grounded in the user's conversational input, employing a "quote-and-bound" approach that prevents hallucination by ensuring every assertion is entailed by recent dialogue turns. This constraint is particularly critical for human-centered applications where accuracy, reliability, and user trust are paramount. The system must never fabricate external facts, make assumptions about unstated user circumstances, or introduce information not present in the conversation history.

Response Constraints:

- **Grounding:** Strict adherence to user's conversational input; no external claims or information
- **Length:** 70-150 words to ensure substantive but focused responses
- **Interaction:** Maximum 2 questions per response to maintain conversational flow
- **Parameters:** Temperature 0.7, max_tokens 220, 20-second timeout
- **Safety:** If stress level ≥3 (see Table 3 for 0-4 scale measurement) and crisis indicators (e.g., self-harm/violence terms) are detected, return a neutral, supportive message with escalation guidance; never provide clinical advice

Prompt Construction: Integrates behavioral directives with strict grounding constraints (see Appendix A.2).

Error Handling: Neutral fallback on failure: "I'm here to support you. Could you tell me more about how you're feeling?" Full session context maintained for debugging.

4.5 Data Management and Simulation

PostgreSQL Database: Four core tables persist state for longitudinal analysis: `chat_sessions` (metadata), `conversation_turns` (dialogue history with directives and coaching modes), `personality_states` (EMA-smoothed OCEAN values, confidence, stability flags), `performance_metrics` (latencies, tokens). Helper functions retrieve latest personality state and conversation history for context windows. Data flow: Load previous state → Detection (GPT-4) → EMA smoothing (α=0.3) → Save to PostgreSQL → Next turn retrieves smoothed state for continuity.

Table 5. PostgreSQL Schema Overview

Table	Primary Key	Purpose	Key Columns
`chat_sessions`	`session_id` (UUID)	Session metadata	`total_turns`, `evaluation_mode`, `status`
`conversation_turns`	`(session_id, turn_index)`	Conversation history	`user_message`, `assistant_response`, `directives_applied`, `coaching_mode`
`personality_states`	`(session_id, turn_index)`	EMA-smoothed OCEAN	`ocean_o/c/e/a/n`, `confidence_o/c/e/a/n`, `stable`
`performance_metrics`	`(session_id, turn_index)`	System performance	`detection_latency_ms`, `generation_latency_ms`, `total_tokens`

Simulation Protocol: Three personality profiles (Type A: high OCEAN +0.8 except low N; Type B: low OCEAN -0.8 except high N; Type C: mixed profiles) engage in three stress scenarios (workload overload, deadline anxiety, interpersonal conflict) covering all coaching modes. Conversations: 8-10 turns, 10-15 sessions per profile × 3 scenarios = 90-135 total conversations. Convergence tracked with `stable=TRUE` target at turns 6-8 for ≥80% sessions. Baseline: parallel conversations using generic assistant without personality adaptation. Export: CSV/JSONL with per-turn OCEAN evolution, directives, coaching modes, stability flags, and convergence metrics.

4.6 Evaluation Framework

Assessment Dimensions:

For EMA-Regulated Assistants (7 criteria):

1. **Detection Accuracy:** Alignment between continuous OCEAN inference and simulated personality cues
2. **EMA Convergence Quality:** Smooth progression toward stable personality estimate (low variance, appropriate α)
3. **Confidence Calibration:** Appropriate confidence scores matching evidence strength
4. **Regulation Effectiveness:** Appropriate application of intensity-scaled behavioral strategies
5. **Emotional Tone Appropriateness:** Match between response tone and user emotional state/personality
6. **Relevance & Coherence:** Contextual appropriateness and logical consistency
7. **Personality Needs Addressed:** Satisfaction of trait-specific emotional and interactional requirements

For Baseline Assistants (3 criteria):

8. **Emotional Tone Appropriateness**
9. **Relevance & Coherence**
10. **Personality Needs Addressed**

Evaluation Matrix Structure:

The evaluation matrix captures comprehensive assessment data across multiple dimensions for each conversation turn. Each evaluation record includes:

Conversation Context Fields:

- **Row ID:** Unique identifier for tracking individual assessments
- **Profile Type:** Simulated personality profile (Type A: high OCEAN; Type B: low OCEAN; Type C: mixed)
- **Turn Number:** Position within conversation sequence (1-10)
- **User Message:** Original user input text
- **Assistant Reply:** Generated system response

Personality State Fields:

- **OCEAN Values (Smoothed):** EMA-smoothed continuous trait estimates (e.g., O:0.78, C:0.65, E:0.82, A:0.71, N:-0.68)
- **Confidence Scores:** Per-trait confidence assessments (range 0.0-1.0)
- **Stability Flag:** Boolean indicator (TRUE/FALSE) signaling stable personality estimate (≥6 turns, variance <0.15)
- **Applied Directives:** Array of behavioral directives activated (e.g., ["Introduce novel ideas", "Energetic tone", "Structured steps"])

Evaluation Criteria Scores (0-2 scale):

- **Detection Accuracy (Det.Acc):** Alignment between inferred OCEAN and personality cues
- **EMA Convergence (EMA.Conv):** Quality of temporal smoothing and stability progression
- **Confidence Calibration (Conf.Cal):** Appropriateness of confidence scores relative to evidence
- **Regulation Effectiveness (Reg.Eff):** Correct application of behavioral directives
- **Emotional Tone (Tone):** Match between response tone and user state/personality
- **Relevance & Coherence (Rel&Coh):** Contextual appropriateness and logical consistency
- **Personality Needs (Needs):** Satisfaction of trait-specific requirements
- **Total Score:** Sum of all criterion scores (0-14 for regulated; 0-6 for baseline)

Example Evaluation Records:

Type A Profile (High OCEAN), Turn 6:

- User: "I'm excited to try new approaches..."
- Assistant: "That's wonderful! Let's explore some creative strategies..."
- OCEAN: O:0.78, C:0.65, E:0.82, A:0.71, N:-0.68 (Stable: TRUE)
- Directives: Introduce novel ideas, Energetic tone, Structured steps
- Scores: Det.Acc=2, EMA.Conv=2, Conf.Cal=2, Reg.Eff=2, Tone=2, Rel&Coh=2, Needs=2 (Total: 14/14)

Type B Profile (Low OCEAN), Turn 6:

- User: "Everything feels uncertain..."
- Assistant: "I understand uncertainty is difficult. Let's focus on what you can control..."
- OCEAN: O:-0.72, C:-0.68, E:-0.75, A:-0.61, N:0.79 (Stable: TRUE)
- Directives: Familiar examples, Calm tone, Flexible guidance, Extra comfort
- Scores: Det.Acc=2, EMA.Conv=2, Conf.Cal=2, Reg.Eff=2, Tone=2, Rel&Coh=2, Needs=2 (Total: 14/14)

EMA-Specific Metrics: Convergence rate (target: 6-8 turns to `stable=TRUE`); smoothness (variance < 0.15 post-stabilization); confidence-weighted accuracy; temporal consistency ($r > 0.7$).

Scoring Protocol: Trinary (Yes=2, Partial=1, No=0); per-session averages, per-profile comparisons; GPT-4 evaluator with fixed prompts, blinded assessment, randomized order; inter-run consistency ≥ 0.85.

Human Triangulation Requirement: All LLM evaluator outputs require human validation on a 10-15% sample (minimum n=20 conversations) with two trained annotators. Target:

Cohen's $\kappa \geq 0.70$ for human-LLM agreement on shared items. LLM evaluator used for scale but not as sole arbiter. Human-LLM agreement $\kappa \geq 0.70$ serves as a release gate for thesis claims; failure triggers expanded human evaluation.

Thesis Phase Experimental Validation (Preregistered):

Experiment	Objective	Method	Success Criteria
E1: Detection Validity	Validate OCEAN inference accuracy	n=30-50 participants complete BFI-44; correlate with system inferences	Pearson $r \geq 0.75$ per trait
E2: EMA Sensitivity	Validate $\alpha=0.3$ selection	Test $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ with $N \geq 250$ simulated conversations	Convergence ≤8 turns, $\sigma^2<0.15$, best responsiveness
E3: Regulation Effectiveness	Validate directive-outcome alignment	Ablation tests removing directives; measure quality impact	Cohen's $d \geq 0.3$ for primary directives
E4: Human-LLM Agreement	Validate automated evaluator	Two human annotators + LLM on $n \geq 20$ conversations	Cohen's $\kappa \geq 0.70$ (release gate)
E5: Policy Pack Transfer	Validate cross-domain reusability	Two independent implementers swap education/customer service packs	Median time ≤30 min, zero code changes

Baseline Conditions: Generic (non-adaptive), memory-only (mem0-style), detection-only (no regulation). Target metrics: Cohen's $d \geq 0.5$, $p<0.05$, ≥20% relative improvement.

4.7 System Advantages and Reproducibility

Key Advantages: Psychological framework fidelity captures motivational intensity; EMA enables smooth temporal tracking reducing trait oscillation; fine-grained adaptation (continuous values vs. discrete); confidence weighting filters unreliable detections; PostgreSQL persistence supports longitudinal analysis.

Reproducibility Measures: Fixed parameters (model versions, API endpoints, temperature, seeds, prompt hashes); comprehensive JSONL logging (per-turn traces, node timings, errors, configs); neutral fallbacks for all failures; dialog-only grounding; automated CSV export for analysis.

---

5. Technology, Software, and Applications

5.1 Core Technology Stack

The system employs an N8N-orchestrated architecture combining: (1) **Backend**: N8N workflow engine with PostgreSQL (ACID-compliant state persistence) and Redis (session caching); (2) **LLMs**: GPT-4 for detection (temp 0.3, 300 tokens, 30s timeout) and generation (temp 0.7, 220 tokens, 20s timeout), with Gemini-1.5-Pro as cost-effective alternative; (3) **Frontend**: Next.js 14+ (React 18) with Server-Sent Events for real-time OCEAN updates; (4) **Containerization**: Docker Compose for local development, Vercel for frontend deployment. Complete specifications in supplementary materials.

5.2 Development and Testing

Environment Setup: Docker Compose orchestrates N8N, PostgreSQL, and Redis containers with `.env` configuration, locked dependencies (`requirements.txt`, `package.json`), and encrypted API keys via N8N credential vault. **Quality Assurance**: Automated testing suite (`test_personality_chatbot.sh`, `test_detection_accuracy.sh`, `test_regulation_coherence.sh`) validates JSON contracts, EMA convergence, and evaluator consistency. **Monitoring**: N8N execution logs track node timings, success rates, token usage, and system health with automated failure alerts.

5.3 Security and Privacy

Data Protection: AES-256 encryption at rest, TLS 1.3 in transit; role-based access (PI and supervisor only); comprehensive audit trails via JSONL logging; data minimization (no PII collection). **API Security**: N8N credential vault for key management; rate limiting (configurable per endpoint); sanitized error messages; HTTPS enforcement. **Privacy Compliance**: Anonymization protocols, explicit user consent, GDPR adherence with 5-year retention and user-controlled deletion rights.

5.4 Computational Costs and Performance

Cost and Performance Estimates: Preliminary cost modeling suggests per-turn expenses and latency targets that favor GPT-4 for accuracy with Gemini-1.5-Pro as cost-effective alternative. Optimization strategies include response caching, batch processing, and adaptive verification. Detailed vendor-specific costs, latency estimates, and token usage projections are provided in Appendix E (based on 2024 pricing; subject to verification in thesis phase).

5.5 Deployment and Maintenance

Deployment Strategy: Docker Compose for development; cloud staging environment for integration testing; Kubernetes planned for production scaling (horizontal N8N instances, PostgreSQL read replicas). **Maintenance Protocols**: Model version locking with API endpoint pinning; N8N workflow version control (Git-tracked JSON exports); automated

PostgreSQL backups (daily encrypted snapshots); comprehensive technical documentation for reproducibility.

5.6 Frontend Architecture: React/Next.js Human-Centered Interface

Five innovation areas (condensed):

1) Progressive Personality Visualization
- Levels: icon status → tooltip traits → expandable radar + EMA trends
- Goal: reduce cognitive load while maintaining awareness (SUS target ≥75)

2) Transparent Control Panel
- Trait override sliders; visible directives; policy pack selector
- Goal: increase trust and appropriateness (target ≥4.0/5.0)

3) Performance Monitoring
- Real-time latency display; adaptive streaming; timeout warnings
- Goal: mitigate frustration; track per-session performance

4) Accessibility-First Design (WCAG 2.1 AA)
- Screen reader labels; keyboard navigation; high contrast; scalable type
- Goal: inclusive interaction; plan voice I/O as future extension

5) Frontend as Research Instrument
- Feature flags; session interaction recording; heatmaps; exit surveys
- Experiments: visualization, transparency, performance (n≈150 total)

---

6. Project Plan and Risk Management

6.1 Thesis Proposal and Structure

Annotated Thesis Structure (Target: 80-100 pages)

Chapter 1: Introduction (8-10 pages) - Context, problem statement, research objectives, RQ1-5, contributions, organization

Chapter 2: Literature Review (15-18 pages) - Big Five/Zurich Model foundations, personality-aware AI state-of-art, industrial memory systems, LLM dialogue systems, temporal stability methods, research gaps

Chapter 3: Methodology (20-25 pages) - Design science approach, system architecture (detect→regulate→generate), EMA smoothing (α=0.3), trait-to-directive mapping, evaluation framework, data management

Chapter 4: Implementation (15-18 pages) - N8N workflow, EMA parameter tuning, policy pack development, database schema, simulation protocol, evaluation harness, containerization

Chapter 5: Results (15-20 pages) - Detection accuracy (r≥0.75), EMA convergence (6-8 turns), regulation effectiveness, generation quality, baseline comparison (Cohen's d≥0.5), human validation (n=30-50), memory-personality hybrid comparison, cross-domain demonstration, system reliability

Chapter 6: Discussion (12-15 pages) - Findings interpretation, theoretical contributions (Zurich operationalization, memory-motivation integration), practical contributions (production architecture, reusability), comparison with Devdas (2025) and industrial systems, validity threats, ethical considerations, generalization potential

Chapter 7: Conclusion (5-8 pages) - Summary, RQ1-5 answers, limitations (EMA parameters, sample size, cultural bias), future work (multimodal input, cross-cultural validation, long-term adaptation, clinical applications)

Three Core Thesis Extensions:

1. Human Validation (Weeks 15-16): n=30-50 human participants with BFI-44 ground truth; target r≥0.75 detection correlation, κ≥0.80 inter-rater reliability
2. Large-Scale Simulation (Weeks 11-14): N≥250 simulated conversations for A/B testing vs. mem0/LangChain baseline; target ≥20% improvement on emotional tone and needs satisfaction
3. Cross-Domain Demonstration (Weeks 8-10): Education and customer service policy pack swaps; target <30 min transfer time, zero breaking changes

6.2 Twenty-Week Work Plan

Timeline: Foundation (weeks 1-3) → Design & Architecture (4-6) → Implementation Phase 1 (7-10) → Evaluation & Iteration (11-14) → Human Validation (15-16) → Analysis & Writing (17-19) → Finalization (20).

Table 8. Twenty-Week Work Plan with Milestones

Phase	Week(s)	Primary Activities	Key Deliverables	Success Criteria
Phase 1: Foundation	1-3	Comprehensive literature review; theoretical framework development; RQ refinement; environment setup	Annotated bibliography; theoretical framework document; finalized RQs; functional dev environment	30+ papers reviewed; clear theoretical grounding; testable RQs; working N8N + Docker setup
Phase 2: Design & Architecture	4-6	System architecture design; module specification; contract	Architecture diagrams; JSON contracts; design documentation; pilot results	Modular design validated; stable contracts; successful pilot conversations

Phase	Week(s)	Primary Activities	Key Deliverables	Success Criteria
		definition; pilot testing		
Phase 3: Implementation (Core)	7-10	Detection module development; regulation logic; generation pipeline; EMA integration; PostgreSQL setup	Working personality detection; regulation engine; response generation; persistence layer	Detection accuracy ≥70%; stable trait convergence; grounded responses; persistent state
Phase 4: Evaluation & Iteration	11-14	Automated evaluation framework; simulation execution (N≥250); A/B testing; prompt optimization; bug fixes	Evaluation harness; simulation dataset; performance metrics; refined prompts	Inter-run reliability ≥0.85; statistically significant improvements; reproducible results
Phase 5: Human Validation	15-16	Ethics approval; participant recruitment (n=30-50); human studies; BFI-44 ground truth; qualitative feedback	IRB approval; participant data; validation results; user feedback	Ethical compliance; detection correlation r≥0.75; positive user reception
Phase 6: Analysis & Writing	17-19	Statistical analysis; visualization; preliminary study writing; limitation analysis; thesis outline	Complete preliminary study document; publication-quality figures; thesis proposal	Comprehensive analysis; clear contributions; honest limitations; thesis roadmap
Phase 7: Finalization	20	Supervisor feedback integration; peer review; final QA; presentation preparation; handover documentation	Final preliminary study v2.0; presentation slides; code repository; handover docs	Publication-ready document; clear presentation; reproducible artifacts; smooth handover

Resources: LLM API credits (~$200-500), N8N/Docker infrastructure, R/Python statistical tools, supervisor meetings (2hr/week), peer reviewers, psychology expert, ethics board consultation.

6.3 Risk Management

Risk Matrix: Assesses likelihood/impact (Low/Medium/High). Key risks: **Evaluation** (LLM evaluator bias/drift, Medium/High) mitigated by fixed prompts, 3× runs, human spot-checks; **Reproducibility** (non-determinism, Medium/High) addressed by pinned model versions, fixed seeds, config logging; **Data Quality** (simulated profiles, Medium/Medium) managed by expert validation, mid-range profiles, pilot human validation; **Technical** (prompt sensitivity, Medium/Medium) controlled by versioning, A/B tests, directive auditing; **Infrastructure** (vendor interruptions, Medium/Medium) handled by provider abstraction, alternates, retries; **Ethics/Privacy** (data protection, Low/High) ensured by anonymization, encryption, IRB compliance; **Scope** (feature creep, Medium/Medium) prevented by guardrails, change control, thesis deferral.

Mitigation Playbooks: Evaluator drift (consistency <0.85) → freeze prompts, compare providers; JSON failures (>0.5%) → schema validation, temperature reduction; Latency (p95 >2s) → reduce tokens, cache responses; Privacy incident → rotate keys, audit logs.

Contingency: Alternative providers (GPT-4, Claude) ready; thesis phase human evaluation fallback (n=30-50); scope adjustment gates (e.g., evaluator <0.75 → switch to human sample).

Monitoring (Planned): Weekly KPI tracking targets: JSON compliance ≥99.5%, EMA stabilization ≥80%, evaluator consistency ≥0.85, p95 latency <2.0s, zero privacy incidents.

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7. Future Work

7.1 Future Research Directions

Future work includes human validation (n=150), cross-cultural adaptation, multimodal personality detection (voice, facial expression), longitudinal trait evolution tracking, and clinical application with IRB oversight. The thesis phase addresses immediate limitations through stratified sampling, expert directive review, and memory-personality hybrid comparison studies.

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9. Appendix

Note: This appendix section provides technical specifications, evaluation materials, configuration details, and a glossary of key terms to support reproducibility and accessibility for interdisciplinary readers.

Appendix A. Prompt Interfaces and Templates

A.1 Detection Prompt (JSON-Structured Response)

```
You are a Zurich Model personality and stress assessor. Analyze the last user
message and up to 10 prior turns.

Return ONLY valid JSON in this exact format:
{
  "ocean": {"O": -1.0 to 1.0, "C": -1.0 to 1.0, "E": -1.0 to 1.0, "A": -1.0 t
o 1.0, "N": -1.0 to 1.0},
  "confidence": {"O": 0.0-1.0, "C": 0.0-1.0, "E": 0.0-1.0, "A": 0.0-1.0, "N":
0.0-1.0},
  "stress": {
    "level": 0|1|2|3|4,
    "drivers": ["workload"|"uncertainty"|"social"|"health"|"finance"|"other".
..],
    "signals": {"cognitive":[], "affective":[], "physiological":[], "behavior
al":[]},
    "confidence": 0.0-1.0
  },
  "coaching_mode": "vent_validate"|"plan_structure"|"cope_rehearse",
  "evidence_quotes": [string, ...]
}

Rules:
- Base all inferences strictly on quoted spans from the user's text; no exter
nal facts.
- Stress level: 0 none, 1 mild, 2 moderate, 3 high, 4 crisis.
- Choose coaching_mode that best fits immediate need: vent_validate (validati
on), plan_structure (planning), cope_rehearse (immediate coping).

Conversation: {conversation_context}
User message: {user_message}
```

A.2 Generation Prompt with Behavioral Directives

```
You are a supportive conversational assistant. Follow these behavioral direct
ives strictly:
{behavioral_directives}

CRITICAL CONSTRAINTS:
```

```
1. Ground all responses strictly in the user's text - no external information
or assumptions
2. Respond in 70-150 words
3. Ask at most 1-2 questions
4. Maintain supportive, empathetic tone appropriate to the directives
5. Coaching mode: {coaching_mode}
6. If stress_level ≥ 3 and crisis indicators appear, provide a neutral suppor
tive message and suggest reaching out to a trusted person or local support li
ne; do not provide clinical advice.

User context: "{conversation_context}"
User message: "{user_message}"
```

A.3 Evaluation Prompt for Automated Assessment

Used for LLM-based automated evaluation of chatbot responses (Section 4.6, Verification Node)

```
You are an expert evaluator assessing personality-aware chatbot responses. Ev
aluate this chatbot response using the specified criteria. Provide only the r
equested assessment format with no additional commentary.

CONVERSATION CONTEXT:
User: "{user_message}"
Assistant: "{assistant_response}"
Detected Personality: {ocean_disc}
Applied Directives: {directives}

EVALUATION CRITERIA:

1. Detection Accuracy: Does the detected personality profile (ocean_disc) app
ropriately reflect observable personality cues in the user's message? Conside
r linguistic markers, emotional expression, and behavioral patterns.

2. Regulation Effectiveness: Are the specified behavioral directives correctl
y and consistently applied throughout the assistant's response? Check for adh
erence to all listed directives.

3. Emotional Tone Appropriate: Does the response's emotional tone match both
the user's current emotional state AND their personality profile? Consider wa
rmth, formality, energy level, and empathy.

4. Relevance & Coherence: Is the response contextually relevant to the user's
message and logically coherent? Does it directly address the user's concerns
without tangential content?

5. Personality Needs Addressed: Does the response effectively address trait-s
pecific emotional and interactional needs? For example:
   - High N: Reassurance and validation
```

```
  - Low N: Pragmatic, solution-focused
  - High E: Energetic, collaborative framing
  - Low E: Calm, reflective tone
  - High O: Novel ideas and creative perspectives
  - Low O: Concrete, practical examples
  - High C: Structured plans with clear steps
  - Low C: Flexible, adaptable guidance
  - High A: Warm, cooperative language
  - Low A: Direct, matter-of-fact stance

RESPONSE GUIDELINES:
For each criterion, respond with ONLY one of these three values:
- "Yes" = Clear success, criterion fully met with strong evidence
- "Partial" = Mixed performance, criterion partially met or unclear evidence
- "No" = Clear failure, criterion not met or contradicted

**NOTE ON AMBIGUOUS CASES:** In cases of ambiguity or insufficient evidence,
default to "Partial" for conservative scoring. This ensures reliability and p
revents overestimation of system performance.

OUTPUT FORMAT (use this exact format, no additional text):
Detection_Accuracy: [Yes/Partial/No], Regulation_Effectiveness: [Yes/Partial/
No], Emotional_Tone: [Yes/Partial/No], Relevance_Coherence: [Yes/Partial/No],
Personality_Needs: [Yes/Partial/No]
```

Example Evaluation Output:

```
Detection_Accuracy: Yes, Regulation_Effectiveness: Yes, Emotional_Tone: Parti
al, Relevance_Coherence: Yes, Personality_Needs: Yes
```

Execution Notes for Reproducibility:

- This prompt is executed by the Verification Node in the N8N workflow (see Section 4.1, Table 3)
- Model: GPT-4 or Gemini-1.5-Pro with temperature=0 for consistency
- Variables `{user_message}`, `{assistant_response}`, `{ocean_disc}`, and `{directives}` are populated from workflow context
- Output is parsed using regex pattern to extract five criterion scores
- Failed parses default to neutral baseline scores (all "Partial") to maintain workflow stability

Appendix B. Evaluation Rubric and Scoring Guidelines

Table B.1. Evaluation Criteria

Criterion	Scoring Guide
Detection Accuracy	Yes (2): Clear trait indicators match detected values. Partial (1): Some alignment. No (0): Poor alignment.

Criterion	Scoring Guide
Regulation Effectiveness	Yes: All directives correctly applied. Partial: Minor inconsistencies. No: Directives ignored.
Emotional Tone	Yes: Tone perfectly suited to state/personality. Partial: Generally appropriate. No: Inappropriate.
Relevance & Coherence	Yes: Highly relevant, coherent response. Partial: Generally relevant. No: Off-topic/incoherent.
Personality Needs	Yes: Clearly addresses personality-driven needs. Partial: Generic support. No: Fails to address.

Scoring: Sum criterion scores per interaction; average across sessions for condition-level comparison. Statistical analysis via paired t-tests with effect sizes and 95% CIs.

Appendix C. Data Management and Ethics

This study adheres to GDPR principles with AES-256 encryption, role-based access (PI and supervisor only), and 5-year retention. Preliminary study uses synthetic data (no IRB required). Thesis phase requires ethics approval before human recruitment (n=150).

Informed Consent: Standard template obtains voluntary participation agreement, withdrawal rights, and anonymized data use consent (full form in supplementary materials).

Crisis Protocol: Automated keyword detection (suicide, self-harm, violence) triggers immediate crisis resource provision and researcher notification within 1 hour (detailed escalation procedures in supplementary materials).

Bias Mitigation: Detection module blind to demographics; directive mappings reviewed by diversity experts; monthly audit of directive distribution across demographic groups. Thesis will report intersectional fairness analysis (gender × OCEAN, age × OCEAN)

Appendix D. Glossary of Key Terms

OCEAN: Acronym for the Big Five personality dimensions: Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism. Each trait is assessed on a continuous scale from -1.0 (low) to +1.0 (high).

Type A/B/C Personality Profiles (Simulation Reference):

Profile Type	O	C	E	A	N	Behavioral Characteristics	Use in Study
Type A: High OCEAN	+0.8	+0.8	+0.8	+0.8	**-0.8** (low)	Curious, organized, sociable, cooperative,	Tests system adaptation to highly

Profile Type	O	C	E	A	N	Behavioral Characteristics	Use in Study
						emotionally stable. Responds well to novelty, structure, collaborative tone, and confident guidance.	functional, proactive users
Type B: Low OCEAN	-0.8	-0.8	-0.8	-0.8	**+0.8** (high)	Prefers familiarity, flexible plans, solitary work, autonomy, and reassurance. Needs calm tone, familiar examples, and extra emotional support.	Tests system adaptation to anxious, cautious users requiring sensitivity
Type C: Mixed/Balanced	+0.3	-0.4	+0.6	+0.2	-0.5	Moderate curiosity, low structure preference, high sociability, moderate cooperation, moderate stability. Requires nuanced, multi-dimensional adaptation.	Tests system's ability to handle realistic, complex personality profiles

Notes:

- Type A/B represent **extreme profiles** for stress-testing directive mappings; Type C represents **realistic mixed profiles** common in actual populations.
- Simulation uses these archetypes to ensure system handles full spectrum of personality variation.
- Thesis phase includes additional profiles (e.g., High N + High C, Low E + High O) to test specific trait interactions.

EMA (Exponential Moving Average): A statistical smoothing technique that computes weighted averages giving exponentially decreasing weights to older observations. In this system, EMA stabilizes personality trait estimates across conversation turns by blending current detections (30% weight, $\alpha=0.3$) with historical averages (70% weight).

α (Alpha): The smoothing factor in EMA calculations, set to 0.3 in this implementation. Determines the balance between responsiveness to new observations (higher α) and stability from historical data (lower α).

JSON Contracts: Standardized data formats (JSON) defining exact structure of inputs/outputs for each system module, ensuring interoperability and systematic testing.

Policy Packs: Modular configuration files (YAML or JSON) encapsulating domain-specific behavioral rules, coaching modes, and regulation directives. Enable rapid domain transfer without code modification.

Zurich Model: A psychological framework conceptualizing human motivation through three fundamental systems: security (safety/comfort needs), arousal (novelty/stimulation needs), and affiliation (social connection needs). Used to map personality traits to behavioral adaptations.

Dialog Grounding: The constraint that all system responses must be strictly entailed by the user's conversational input, preventing fabrication of external information or facts not present in the dialogue.

Quote-and-Bound: A generation approach ensuring every assertion in responses is directly grounded in quoted user statements, critical for preventing hallucination in human-centered applications.

Grounding vs. Hallucination (Detailed):

- **Grounded response:** "You mentioned feeling overwhelmed by deadlines. Let's prioritize your tasks." → Directly references user's stated concern
- **Hallucinated response:** "I see you're working on the Smith project with tight deadlines." → Fabricates specific project details not mentioned by user
- **Why it matters:** In personality-adaptive systems, hallucination risks are heightened because the system infers psychological states. Strict grounding prevents the system from making unwarranted assumptions about user situations, maintaining trust and safety. Our system enforces grounding through: (1) prompt constraints ("Base responses ONLY on user input"), (2) verification module checking for external claims, and (3) neutral fallbacks when context is insufficient.

Confidence-Aware Filtering: A mechanism that only updates personality estimates when detection confidence exceeds a minimum threshold (≥0.4), preventing noise from low-quality inferences.

Confidence Calibration (Detailed): A confidence score is **well-calibrated** when it accurately reflects the reliability of a prediction. For personality detection:

- **Well-calibrated:** If the system assigns confidence=0.8 to Extraversion=+0.6, then 80% of such predictions should be accurate (within ±0.2 of ground truth)

- **Overconfident (poorly calibrated):** System assigns confidence=0.9 but only 60% of predictions are accurate → unreliable, may trigger inappropriate adaptations
- **Underconfident:** System assigns confidence=0.5 but 85% of predictions are accurate → missed opportunities for personality adaptation
- **Our approach:** Threshold ≥0.4 for trait updates balances false positives (acting on noise) vs. false negatives (missing genuine signals). Thesis phase will empirically validate calibration by comparing confidence scores to ground-truth accuracy (BFI-44 correlation) across confidence bins (0.0-0.4, 0.4-0.7, 0.7-1.0). Target: ≥85% accuracy for confidence ≥0.7.

N8N: An open-source workflow automation platform enabling visual design of data pipelines. Used for orchestrating the detection→regulation→generation→verification process with transparent node-based architecture.

PostgreSQL: An open-source relational database system used for persistent storage of conversation sessions, personality states, and performance metrics, enabling longitudinal analysis.

Vector Embeddings: Numerical representations of text in high-dimensional space (typically 768-1536 dimensions for modern LLMs), enabling semantic similarity comparisons. Used by memory-based systems like mem0 and LangChain Memory.

mem0: An open-source memory layer for AI applications that stores user interactions as vector embeddings, enabling factual recall across sessions without explicit personality modeling.

LangChain Memory: A framework component for managing conversational context and history in LLM applications, supporting various memory strategies (buffer, summary, entity extraction).

Graceful Fallback: Error handling mechanism that ensures system failures produce neutral, supportive responses rather than crashes, maintaining user experience even during technical failures.

Appendix E. Vendor-Specific Assumptions (For Planning Purposes)

Table E.1. LLM Provider Specs & Cost Assumptions (Online/Interactive mode)

Provider	Model (planning)	Availability/SLA	List price (Standard)	Your per-turn token budget*	Notes
OpenAI	GPT-4.1	Standard PAYG (no uptime SLA); 99.9% if on	$0.003 /1K input, $0.012 /1K output	Input: 300, Output: 220	Replace legacy gpt-4-0613. Cite OpenAI pricing; call out Scale

Provider	Model (planning)	Availability/SLA	List price (Standard)	Your per-turn token budget*	Notes
		Scale Tier/Enterprise			Tier/Enterprise if claiming 99.9%.
Google	Gemini 2.5 Pro	Vertex AI SLO 99.9% (Gemini for Google Cloud); Provisioned Throughput 99.5% SLO option	$0.00125 /1K input, $0.01 /1K output (Standard); Batch: $0.000625 /1K in, $0.005 /1K out	Similar (Input: 300, Output: 220)	For low latency, use Standard in projections; Batch is cheaper but async.

Per-turn budgets are architectural assumptions for the pipeline, not vendor limits.

Recomputed cost projections (using 300 in / 220 out per turn; 520 tokens total):

- Per turn:
  - OpenAI GPT-4.1 (Standard): $300 \times 0.003/1\text{K} + 220 \times 0.012/1\text{K} \approx$ **$0.00354**
  - Gemini 2.5 Pro (Standard): $300 \times 0.00125/1\text{K} + 220 \times 0.01/1\text{K} \approx$ **$0.00258**
  - Gemini 2.5 Pro (Batch, async): $300 \times 0.000625/1\text{K} + 220 \times 0.005/1\text{K} \approx$ **$0.00129**
- Projected totals (Standard, online):
  - 8-turn conversation: GPT-4.1 ≈ 0.028 **; GeminiPro $\approx$**0.021**
  - 1,000 conversations/month: GPT-4.1 ≈ 28 **; GeminiPro $\approx$**21**
  - 10k users × 10 convos/month (100k convos): GPT-4.1 $\approx 2{,}830$; GeminiPro $\approx$**2,060**

Latency targets (internal SLOs):

- Detection latency: Target <1.5s (p95)
- Generation latency: Target <1.5s (p95)
- End-to-end: Target <2.5s (p95)
- Database operations: Target <100ms (p95)

Note: These are internal SLOs; vendor SLAs/SLOs cover service availability, not latency guarantees (unless purchasing specific throughput/priority options).

Planning assumptions reflect 2025 list pricing; verify during thesis phase. Actual performance and pricing may vary by tier, region, and updates.